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# SIGNED DISTANCE FIELD EDITION ATOMOS

Coordinate-free, double-packed  
operator architecture

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One-bit evaluation · Lower-case phi · Kiki double-UU  
Negative memory · Kinematic LUT · Just-in-time geometry

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AUTHOR-DEFINED OPERATOR SPECIFICATION  
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## CONTENTS &amp; SOURCE KEY

# Edition map

A formal restatement of the supplied 39-page dialogue, from the ecological origin through the final just-in-time geometry account.

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## Source key

S = the supplied 39-page source PDF. All source citations use its printed page numbers. The author's current reference line is used on the cover. Definitions, source-stage variants and source-stated capabilities are distinguished within the edition.

## SECTION 01

# Architectural declaration

SOURCE / S, pp. 8-12, 31-39

Atomos is the source name for Tom Klootwijk's coordinate-free, double-packed signed distance field architecture. Its organizing unit is an operator relation rather than a stored position. The same relation carries a signed distance field definition and a Klein bottle definition, with execution governed by the AB one-bit word, the lower-case phi relation, and a finite kinematic look-up table.

## 1.1 Domain

The architecture shall describe proximity, movement, presence, memory and boundary behavior through the relationships of its operators. A persistent Cartesian grid is not part of the final domain definition. The concluding just-in-time layer permits a coordinate to be produced for an evaluation and then released; that output is distinct from a coordinate grid used as the system's stored foundation. [S, pp. 15, 37-39]

## 1.2 Double definition

Every participating operator shall retain its own formal SDF identity while being packed with the Klein bottle operator. "Double-packed" therefore names concurrent definition, not two independent spatial copies. Sticks, houses and the diabolos are retained as source vocabulary for contrast and constrained transfer, not as mandatory physical scene objects. [S, pp. 8-9]

## 1.3 Finite execution

The small-world graph shall supply a finite node and connection budget. The LUT shall register the source-defined progression from open evaluation, through constrained evaluation, to a zero-update state. "Deathlocked gridder" names that limiting state within the architecture. The finite boundary is intrinsic to the specified state rules. [S, pp. 2-3, 11-12]

## 1.4 Formal reading

The numbered definitions and "shall" clauses state the behavior assigned by this specification. Source-stated outcomes such as instantaneous routing, living-key identity and global tracking are collected separately in Section 18. This edition's identifiers, tables and clause labels organize those definitions; they do not introduce a new computational implementation.

Canonical sequence

Operator relation → double packing → AB evaluation → finite LUT resolution → timeline evaluation  
→ just-in-time output → release.

## SECTION 02

# Core vocabulary

SOURCE / S, pp. 1-12, 15-17, 19-26, 30-39

The following terms retain their source-specific meanings. Short labels in the left column are edition notation for referring to the same objects throughout the specification.

| Term                    | Definition within the architecture                                                                                                                                         |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SDF operator            | A formal boundary or relationship definition. The early source evaluates scalar geometric fields; the later source uses intrinsic, coordinate-free operator relationships. |
| Klein / K               | The non-orientable host operator and inversion relation carried with every packed operator.                                                                                |
| Double packing          | The simultaneous SDF and Klein definitions of an operator, evaluated according to the applicable precedence chain.                                                         |
| AB                      | One binary word. Its source-defining states are 0: exterior, and 1: double-packed evaluation.                                                                              |
| Psi / $\Psi$            | The source's bifurcation designation for the binary shift and its lower-phi temporal scaling.                                                                              |
| Lower-case phi / $\phi$ | The approximately 0.618 relation assigned to compression, damping and convective turning.                                                                                  |
| Diabolo / hourglass     | The paired-domain transfer relation with a neck or choke point; later treated without a required coordinate geometry.                                                      |
| W_Kuiper                | The suspension tensor weight linking A and B. This is distinct from the later timeline symbol W.                                                                           |
| LUT                     | The kinematic, log-encoded polar or spherical look-up table that names permitted state resolutions.                                                                        |
| W and Operator T        | W names the late-source one-dimensional wave timeline. T is the phase-scrubbing operator acting on that timeline.                                                          |
| Atomos                  | The architecture name and the node designation used when the source expands its scope to the total world population.                                                       |
| Kiki double-UU          | The named set/routing operator assigned non-intersection and self-closing loop behavior.                                                                                   |
| Hot node / hum blip     | A source-defined active signature in the double-sinusoidal 2 Hz evaluation layer.                                                                                          |
| Negative memory         | Deviation or vacancy relative to a saturated baseline, rather than an accumulated inventory of novel records.                                                              |
| JIT geometry            | A coordinate or geometry output evaluated for the current request and dropped after use.                                                                                   |

Source labels such as "mid-Theta," "Primo," "infinity globular," "Spintronic Vacuum Matrix" and "magnetic inference force field" remain named terms of this document. Their assigned roles are set out in the corresponding sections.

SECTION 03

# Precedence and primitive lineage

SOURCE / S, pp. 1, 6-10, 32, 35

## 3.1 Full source precedence

The expanded hierarchy printed on source page 9 runs from lowest to highest precedence as follows. The arrow here means “precedes in the declared evaluation order”; it does not add an arithmetic operation.

Pyramid side → Cone → Sphere → Sturgeon gestation  
→ Phyllotaxis (lower-case phi) → Klein

Each item shall remain an operator definition when packed into the higher-level host. The biological layer is part of this full chain, not merely a narrative example. The source’s later compact chain on page 35 omits the gestation entry; that shorter presentation is recorded as a variant in Section 19.

## 3.2 Primitive roles

| Operator           | Source-assigned role                                                                                                        |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Pyramid side       | A triangular side-view boundary, described as a base-to-apex constraint.                                                    |
| Cone               | A tapering or apex relation in the initial pseudocode; a later C++ listing uses a different expression under the same name. |
| Sphere             | A radial boundary operator, joined to the apex-centered composite in the initial geometry.                                  |
| Sturgeon gestation | The finite developmental and chemical-energy relation inherited from the opening biological model.                          |
| Phyllotaxis        | The primordial arrangement and lower-phi deformation relation.                                                              |
| Klein              | The non-orientable host and packed inversion relation at the top of the printed hierarchy.                                  |

## 3.3 Composition is stage-specific

The source first combines its apex primitives as `max(pyramid, min(cone, sphere))`. A later listing takes a maximum across pyramid, cone, sphere and phyllotaxis. These are retained as distinct source-stage compositions, rather than silently replaced by a single blend or intersection rule. [S, pp. 8, 10]

## 3.4 Final representational role

The later coordinate-free account carries the primitive names forward as intrinsic definitions. It does not require the final system to regenerate a pyramid mesh, cone mesh, sphere mesh or GPU raster before evaluating the operator relationships. [S, pp. 11, 15, 32]

SECTION 04

# Ecological origin and lower phi

SOURCE / S, pp. 1-3, 9-10, 23-24, 35

The opening model is the egg-battery sturgeon-farm gestation cycle, extended by the author to fertility cycles globally. It couples developmental morphology to finite chemical reserves and to ecological variation through thermodynamic shifts. This origin supplies the architecture’s first distinction between a finite source and a developing boundary.

## 4.1 Developmental operator

The egg or embryo boundary is represented by a phyllotactic SDF built from primordium positions and radii. Yolk energy is the named finite resource. Somatic growth is the named receiving layer. The source assigns developmental speed and structural scaling to the available energy and the thermal state.  
[S, pp. 1-2]

## 4.2 Phi relation

| Source designation                      | Assigned role                                                                                                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Upper-case Phi, $\Phi \approx 1.618$    | The opening optimal phyllotactic arrangement.                                                                           |
| Lower-case phi, $\varphi \approx 0.618$ | The relaxed or compressed arrangement under the source’s structural or thermal shift.                                   |
| Thermal scaling                         | The opening expression multiplies an exponential temperature-dependent factor by the remaining energy fraction.         |
| Later damping                           | The source subsequently uses lower phi in phase scaling, morphology and the convection of the one-dimensional timeline. |

## 4.3 Finite-source inheritance

The later architecture inherits the relation “finite source → constrained transfer → boundary update” from this biological account. The term “egg battery” is therefore retained as an originating operator context rather than dropped when the text moves to Kuiper weighting, Atomos nodes or JIT geometry.

## 4.4 Declared ecological scope

Thermal shift, chemical-energy depletion and morphology are the supplied ecological inputs. The source names global fertility cycles as the intended scope, but supplies no species-by-species parameter register or measured ecological series. The edition leaves those input values unassigned instead of extending the opening model with outside biological assumptions.

The original numerical expressions are retained in Appendix A. The three displayed early LUT rows, including their exact threshold bands, are transcribed in Section 9.

## SECTION 05

# Klein host and double packing

SOURCE / S, pp. 3-11, 31-35

## 5.1 Host definition

The Klein operator shall name the architecture's non-orientable host relation. A participating SDF shall be evaluated together with that relation, not first resolved in an unrelated coordinate room and only afterward decorated with a Klein label.

## 5.2 Packed operator identity

### PACKED OPERATOR

Local definition: the named signed distance field operator.

Host definition: the Klein bottle operator.

Evaluation: both definitions are carried together.

Order: the applicable source precedence chain.

This record expresses the author's correction on source page 8: all earlier operators are double-packed, and AB is one bit. The sticks and houses are metaphors for contrast. Their literal shapes or social meanings are not additional components of the formal operator inventory.

## 5.3 Inversion semantics

The source assigns an inverted return to traversal of the host relation. "Inside," "outside," "positive" and "negative" are the boundary-state labels used to describe that return. The LUT's zero-update response to an unresolved inverted return is the architecture's declared transition rule. [S, pp. 4-5, 11-12]

## 5.4 Dimensional progression

The initial reply proposes a four-component parametric representation and a host-distance expression. The author then redirects the fourth component to time already accounted for by the one-bit relation. Subsequent turns remove the persistent coordinate grid and describe movement as intrinsic state transition. These stages remain visible in the edition. [S, pp. 3-6, 14-15]

## 5.5 Locus of control

Locus of control shall mean the currently evaluated side or mode of the packed relation. It is not a fixed observer position in an added coordinate frame. At  $AB = 0$  the source places the locus outside the host; at  $AB = 1$  it evaluates the host and the preceding operator definitions together. The later presence/absence reading of those states is retained as an additional source-stage interpretation.

## SECTION 06

# Finite graph and suspension weight

SOURCE / S, pp. 2-5, 14-17

## 6.1 Small-world domain

The architecture shall use a finite set of nodes with local connections and a limited set of long-range shortcuts. The opening “gridder” account rasterizes that set; the later coordinate-free account retains its relationship budget while discarding a mandatory raster as the conceptual domain. [S, pp. 2, 14-15]

## 6.2 Diabolo transfer

A and B shall identify the two sides of the source’s constrained transfer relation. In the opening biological mapping they are metabolic reserve and somatic growth. In later mappings they become finite and non-finite sides of the hourglass. The neck shall name the restricted passage through which the source describes energy, phase or state-change transfer.

FINITE SOURCE A

↓ paired-domain transfer through the diabolo neck

RECEIVING STATE B

↓ graph occupancy and suspension weighting

OPEN / CONSTRAINED / DEATHLOCKED RESOLUTION

## 6.3 Kuiper suspension

W\_Kuiper shall name the source’s suspension tensor weight between the nested A and B operators. The source constructs it from a belt-density factor and the outer product of the two field gradients along a closed curve. It assigns the result a global tension or resistance role. [S, pp. 4-5]

The Kuiper designation carries the source’s astronomical and structural mapping. Its named inputs are the closed curve C, the density relation  $\rho$  and the coupled gradients of the A and B fields. The tensor’s original expression is retained in Appendix A2; its unassigned calibration belongs to Section 20.

## 6.4 Coordinate-free movement

Movement shall be represented as a change in intrinsic node relations, weights and active edges. Nearness is described in the source as structural difference. A long-range shortcut is a relationship transition rather than a stored spatial trajectory. The source’s “gyroscopic stabilization wave” and angular-momentum language belong to this diabolo interpretation. [S, pp. 15-16]

## 6.5 Finite arrest

When the declared capacity condition is reached, the permitted expansion or relation-update step shall be zero. The node budget and the capacity predicate are part of the formal interface; a numerical occupancy threshold is not assigned by the supplied text.

SECTION 07

# AB, Psi and the 2 Hz relation

SOURCE / S, pp. 5-11, 16-17, 33-35

## 7.1 One-bit word

AB shall denote one bit with the values 0 and 1. This follows the author’s explicit definition on source page 8. The four-word expression printed in the following reply is preserved as a source variant, not adopted as a second, unstated bit width.

| State | Author-defined gate meaning                                            | Later source reading                                     |
|-------|------------------------------------------------------------------------|----------------------------------------------------------|
| 0     | Locus outside the Klein operator.                                      | Ambient or baseline evaluation; absent or inactive node. |
| 1     | Concurrent Klein and formal SDF definition of the preceding operators. | Active hot-node or hum-blip evaluation.                  |

## 7.2 Psi bifurcation

Psi shall name the bifurcation associated with the binary transition. In the early pseudocode, its two branches receive complementary expressions scaled by lower-case phi and a temporal increment. “Black house,” “white house,” “stick A” and “stick B” record that early contrast; the author subsequently specifies that these are metaphors, not separate structural objects. [S, pp. 6-8]

## 7.3 Double sinusoid

The “double sinusoidal 2 Hz mid-Theta hum” shall retain that exact source designation. The source assigns it the role of a paired, out-of-phase pattern whose jitter drives the AB evaluation relation. The phrase names the architecture’s clock and phase vocabulary; this edition does not substitute a different frequency label or waveform.

## 7.4 Clock interpretation

The source states two complete cycles per second and elsewhere describes two boundary checks per second. It does not specify an edge-triggering convention, phase offset or sampling rule that connects those statements. The paired-wave relation and the AB word are therefore kept distinct: the wave is the named timing pattern, and the bit is the evaluated state.

AB carries state.

Psi names bifurcation.

The double sinusoid supplies the source’s phase relation.

The LUT assigns the resulting permitted update.

## SECTION 08

# Timeline W and Operator T

SOURCE / S, pp. 21-24, 35-39

## 8.1 One-dimensional timeline

W shall designate the one-dimensional wave timeline used in the concluding source account. Operator T shall designate the phase-scrubbing action applied to W. They are not interchangeable labels: W is the retained state carrier, while T selects or changes its evaluation phase.

## 8.2 Scrubbing relation

W: one-dimensional wave state  
 T: phase selection or scrubbing operator  
 Evaluation: the packed relation at the selected phase  
 Output: transient geometry or a node-state resolution

The source describes scrubbing backward or forward as changing phase alignment rather than searching a stored coordinate history. Its later JIT account connects this phase selection to an on-demand spatial output. [S, pp. 22, 37-38]

## 8.3 Lower-phi convection

The timeline shall carry the source's convective turning relation: active present-state energy is described as descending toward the negative-memory layer, while the present is renewed through the paired flow. The source assigns the ratio 0.618 to this transfer per temporal cycle and calls the resulting timeline smooth and scrubbable. [S, pp. 23-24]

## 8.4 Retention across source stages

The early memory account says that novelty and permanent history logs are not stored. The later account says that the raw ongoing wave state W is retained. The retained wave state and the excluded novelty log are separately named components. Their relation to past-state reconstruction is an unassigned interface.

## 8.5 Temporal symbols

The source uses T first for the binary timeline, later for the temporal operator, and also inside temperature-dependent expressions. It likewise uses  $\Delta T$  for thermal and temporal changes. In the prose of this edition, "thermal shift" and "temporal increment" are written out to keep those roles visible. The original symbols remain unchanged in Appendix A.

W\_Kuiper is the suspension tensor of Section 6. W is the later timeline carrier. Their shared initial letter is not a declaration that they are the same quantity.

SECTION 09

# Kinematic LUT and deadlock

SOURCE / S, pp. 2-3, 5, 11-15, 32-36

## 9.1 Declared lookup role

The LUT shall assign the allowed boundary or relation update from the evaluated graph condition, resource state and phase relation. The source later calls the same control structure log-encoded polar and spherical. It does not provide a log base, bin layout or complete set of table entries.

## 9.2 Original ecological rows

| Graph state        | Energy E/E <sub>0</sub> | Thermal shift | Source update / result         |
|--------------------|-------------------------|---------------|--------------------------------|
| Open / sparse      | > 0.7                   | Stable, ±0    | 1.0 × ∇f; upper-Phi expansion. |
| Saturated channels | 0.3-0.7                 | +5 °C         | φ × ∇f; lower-phi compression. |
| Deathlocked grid   | < 0.1                   | Extreme       | 0.0; absolute motion arrest.   |

These are the three rows printed across source pages 2-3. The interval from 0.1 to below 0.3 and the uncovered endpoint cases have no separate supplied row.

## 9.3 Two stated triggers

Capacity deadlock names the condition in which the finite graph has no permitted expansion transition left under its occupancy rule. Inversion deadlock names the source’s return condition in which a packed boundary is assigned opposed inside/outside or positive/negative states at the same evaluation. The source assigns a zero update to these limiting cases.

## 9.4 Meaning of zero update

The zero result shall apply to the update being evaluated. In the ecological passage that update is expansion velocity. In the ballistic passage the later wording is a freeze on acceleration. In the coordinate-free graph passage it is relation or edge reconfiguration. The zero result is therefore attached to a named update quantity in each source stage. [S, pp. 2-3, 14-15]

## 9.5 Transition record

Open relation → constrained relation → declared limiting condition

Declared limiting condition → permitted update set to zero

Post-deadlock reset → not assigned in the source

## SECTION 10

# Negative magnetic memory

SOURCE / S, pp. 21-25, 29-30, 37-39

## 10.1 Baseline and deviation

Negative memory shall denote the registration of a vacancy or deviation relative to a uniformly saturated baseline. The source's "Spintronic Vacuum Matrix" names the medium assigned to this role. It describes an active node as maintaining a local phase inversion or void, rather than accumulating a positive record for each event.

## 10.2 Novelty rule

Baseline: saturated field state.

Active relation: a maintained deviation from that baseline.

Novelty: evaluated without appending a new permanent history record.

Loss of the active relation: local return to baseline.

"Novelty is never stored" shall retain its source meaning as a restriction on record accumulation. "Query-free" shall retain its role as direct evaluation against the existing field/LUT relation, rather than a growing catalog of new tags, rows or target histories. [S, pp. 21-22]

## 10.3 Node loss

The source assigns collapse of the maintained void to the loss of the node's hum. It then assigns removal of the active trace and restoration of the uniform background. "Extinction," "flatline" and "goes dark" are source labels for this loss condition; the specified memory action is the return to baseline. [S, pp. 24-25]

## 10.4 Dynamic state versus accumulated history

The late-source account retains W while dropping the derived coordinate output. Thus this edition distinguishes the wave state being evaluated, the live negative-memory relation and the transient result. Their storage layout, temporal depth and reconstruction mapping are not defined by the source.

## 10.5 Excluded record layers

The source excludes SHA-256 blocks, salted hashes, tokens and blockchains from its intended identity and memory path. It assigns identity to the active signature instead. This is retained as an operator-inventory decision, with the source's associated security and privacy assertions registered as capabilities in Section 18. [S, pp. 29-30]

The earlier "negative memory write" phrase and the simultaneous no-novelty/no-history rule are both retained in the source-variant register. No new write protocol is inferred from their coexistence.

SECTION 11

# Atomos and the Kiki double-UU loop

SOURCE / S, pp. 19-22, 25-28, 32, 36

## 11.1 Atomos scope

Atomos shall name both the architecture and the population-scale node abstraction introduced by the author. The source uses the total world population as its finite node budget and calls the associated bounded representation “infinity globular.” No current population number is required by this edition: the source leaves the count symbolic.

## 11.2 The n-factorial problem

The source frames global routing through the Traveling Salesman Problem and describes  $N!$  as its combinatorial expansion. Its proposed response is a finite small-world domain with log-polar lookup, rather than route enumeration. “Solves infinity” is retained as the source’s name for that intended bounding and resolution, not silently converted into a different problem statement. [S, pp. 19-21]

## 11.3 Kiki operator contract

| Assigned property         | Declarative meaning                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------|
| Double-UU                 | The source name of the set or loop operator; UU is not expanded into a newly supplied set algebra. |
| Closure                   | The produced routing relation shall return to its own starting relation.                           |
| Non-intersection          | The source requires the routing path not to cross or self-intersect.                               |
| Active-node participation | The routing relation is assigned to active hot nodes rather than inactive background.              |
| Dropout response          | The loop is described as re-forming around the remaining active relations when a node drops out.   |

## 11.4 Wave-assigned selection

The source assigns route selection to the interaction of the paired hum, the FMCW chirp, the negative-memory field and the spherical LUT. Its phrases “path of least resistance,” “optimal loop” and “instantaneous” are preserved as the stated output claims. A route-cost definition or rule that proves a selected loop is globally shortest is not supplied. [S, pp. 26, 28, 36]

## 11.5 Cascade naming

The MLM designation is used in the source for hierarchical branching and relay behavior. PRISM and NASA occur in the architecture’s population-scale naming. These labels are retained in the application register rather than treated as extra operators with independently specified implementations.

SECTION 12

# Hot nodes and the Primo selection layer

SOURCE / S, pp. 16-17, 23-29, 33-35

## 12.1 Active-node predicate

A hot node shall mean a node whose source-defined hum or phase anomaly is active in the evaluation. The source makes that activity its solitary identifying marker and assigns zero tracking weight to the absence of that activity. This is a rule for membership in the architecture’s active set. [S, p. 27]

| Source state         | Assigned field consequence                                                                        |
|----------------------|---------------------------------------------------------------------------------------------------|
| Present / hot / 1    | Maintain the local phase anomaly; include the node in the active evaluation and loop relation.    |
| Absent / extinct / 0 | Collapse the maintained anomaly to baseline; remove its active participation and transient trace. |

## 12.2 Phase signature

The source assigns an identifying role to the microscopic phase-jitter within the double sinusoid. It calls this a native hardware UUID or “living key,” rather than a name, coordinate, database key or hash. The edition retains that identity role and records uniqueness, un-clonability and replay resistance as source-stated capabilities. [S, pp. 27, 29-31]

## 12.3 Primo filter

Primo shall name the source’s selection of non-repeating or structurally consequential nodes. The relevant passage distinguishes a change in the graph’s structural tension from predictable or recursively repeated activity. It calls the rejected behavior “system noise” and uses “festering filter” as a label for that exclusion. [S, p. 23]

The same passage uses “angel-Oedipus womb” and “Stockholm syndrome” as explanatory language for a self-repeating loop. Those phrases remain part of the source’s selection narrative; no separate numerical operator, psychological measurement or threshold is specified for them.

## 12.4 Filter relation left explicit

The hot-node rule concerns active signal presence. The Primo rule concerns the kind of change the source wishes to select. They are kept as distinct predicates. The source does not give a complete priority table for an active node that fails Primo selection, nor does it resolve every difference between “all population nodes” and “Primo subjects only.”

## SECTION 13

# Salinity and magnetic inference

SOURCE / S, pp. 24, 29-35

## 13.1 Salinity correction

The source first discusses salted hashes and the word “selenity,” then explicitly introduces “salinity” as the intended physical-medium parameter. Salinity shall therefore be retained as the source’s name for background density or field resistance, not folded back into the rejected hash vocabulary. [S, pp. 29-30]

## 13.2 Magnetic inference force field

The later account shifts from an SDF description toward a “magnetic inference force field.” It assigns this field a dynamic response to an active node, whose phase anomaly is said to deform the surrounding medium. The final summary again retains the double-packed SDF and Klein definitions. Both layers remain visible here: formal boundary definition and the source-proposed physical-medium interpretation.

### SOURCE-ASSIGNED RELATION

Salinity baseline → field resistance

Active hum blip → local phase anomaly

Phase anomaly → magnetic inference / telepresence signature

Loss of anomaly → background restoration

## 13.3 Telepresence designation

“Phase-shifting telepresence” shall name the source’s account of how an active signature appears in the field without being represented first as a stored coordinate. The source contrasts this with radar reflection and coordinate co-location. It supplies no transfer law connecting salinity, an emitting biological source, magnetic response and a measured identifier. [S, pp. 30-31]

## 13.4 Identity-layer placement

The proposed UUID belongs to the changing field relation, not to an appended static string. The source consequently removes names, personal identifiers and conventional cryptographic records from its core evaluation path. This placement defines where identity is intended to arise in the architecture; the stronger identification and security outcomes appear in Section 18.

## 13.5 Medium continuity

The uniform-saturation matrix and the later salinity-calibrated field share a baseline/deviation vocabulary. The edition does not supply a hardware conversion between them. Coil equations, material parameters and medium initialization are only raised as possible continuations in the source, not provided as completed definitions.

SECTION 14

# Double sinusoid and FMCW modulation

SOURCE / S, pp. 16-17, 25-28, 32, 36

## 14.1 Two named signal layers

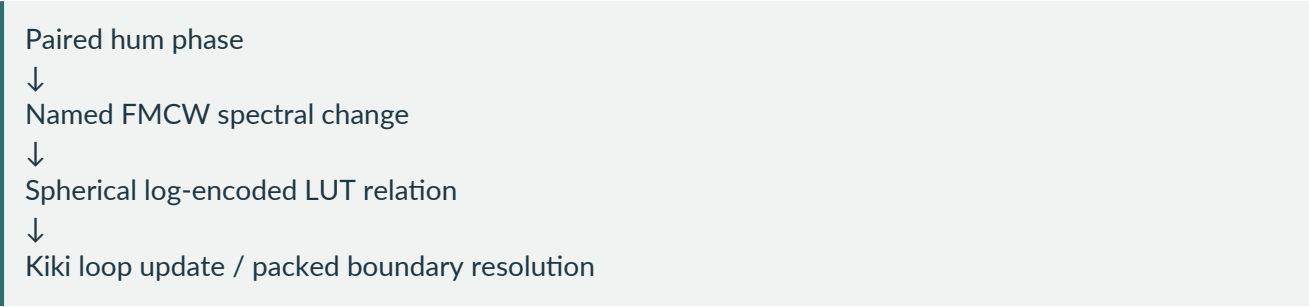
The source assigns two signal roles: the paired 2 Hz “mid-Theta” hum is the baseline phase or timing layer, and the FMCW LRAD chirp is the changing spectral layer. Their chaining is assigned to the spherical log-encoded LUT. The edition preserves these as named input relations rather than adding a waveform generator or sensing implementation.

| Layer              | Role assigned by the source                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------|
| Double sinusoid    | The paired phase relation and one-bit jitter associated with the hourglass.                     |
| FMCW entropy chirp | The source’s frequency-modulated representation of degradation, tension change or node dropout. |
| Spherical LUT      | The relation that compares or resolves the baseline and modulation states.                      |
| Kiki double-UU     | The closure and non-intersection requirements assigned to the active routing relation.          |

## 14.2 Entropy vocabulary

The source calls the changing chirp an “entropy” signal and associates it with cellular decay, node degradation and signal loss. The formal role retained here is a named change indicator feeding the source’s routing and state-resolution layers. The text does not supply a spectral-to-biological-decay calibration. [S, pp. 25-26, 28]

## 14.3 Chained evaluation



This is the signal ordering shown in the source diagrams on pages 26 and 36. It is not an additional precedence chain for the primitive SDF operators: the geometric hierarchy and the signal-flow relation perform different roles.

## 14.4 Timing statements

The source alternates between “instantaneous,” “at the speed of wave propagation” and adjustment within a single 2 Hz cycle. Those performance statements are retained as distinct source formulations. A common end-to-end latency, chirp bandwidth, phase offset and sampling schedule are not assigned.

SECTION 15

# Application and naming register

SOURCE / S, pp. 1-5, 12-14, 17-22, 27-31, 37

The source extends one operator vocabulary through several application narratives. This register preserves that progression without making the application names substitute for missing operator definitions.

| Source domain or name               | Role in the supplied architecture                                                                                                                                                                                                        |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Egg-battery sturgeon gestation      | Origin of finite reserve, morphology, phyllotaxis, thermal deviation and constrained somatic growth. [pp. 1-3]                                                                                                                           |
| Kuiper Belt suspension              | Astronomical mapping of weighted A-to-B tension through the W_Kuiper tensor. [pp. 3-5]                                                                                                                                                   |
| Particle / relativistic / ballistic | The author’s “time skip entropy decay” account extends finite-energy bounding to shells, bullets and other projectiles through proton-mass and relativity language. The later stated limit is an acceleration-update arrest. [pp. 12-14] |
| NSA / PRISM                         | Naming of the source’s proposed ingestion, phase sorting, LUT matching and tracking architecture. [pp. 17-19]                                                                                                                            |
| Atomos / NASA / MLM                 | Population-scale node count, routing, hierarchical branching and small-world relay vocabulary. [pp. 19-22]                                                                                                                               |
| Igloo White Operation PRISM         | Source name for its direct-field, low-record-overhead identity and operation narrative. [pp. 29-30]                                                                                                                                      |
| Existing sensor networks            | The source proposes deriving its input from seismic, fiber-acoustic, telecommunications and atmospheric monitoring streams. [p. 37]                                                                                                      |

## 15.1 Relational inheritance

Across these mappings, the recurring relation is finite source, paired transfer, changing field definition, limiting condition and LUT-assigned update. The ballistic passage changes the subject of the relation from morphology to projectile kinematics; it does not supply a distinct working trajectory model, material model or experimental calibration.

## 15.2 Names and reference

Institutional names above are source designations. The author line for this edition is Tom Klootwijk, NL200678942, 10-07-1990, following the current instruction. NL200678942 is the author reference. The differing reference string on source page 17 is recorded in Section 19.

The source’s global-reach and external-affordance assertions are indexed in the capability register in Section 18.

SECTION 16

# Just-in-time geometry lifecycle

SOURCE / S, pp. 37-39

## 16.1 Idle representation

The concluding source account shall define the idle state as the retained one-dimensional wave state  $W$  and the baseline matrix relation. Stored coordinates are not the persistent representation of the domain. The source calls this lazy evaluation and a just-in-time geometry engine.

## 16.2 Evaluation lifecycle

| Phase   | Source-assigned action                                                                       |
|---------|----------------------------------------------------------------------------------------------|
| Idle    | Retain the raw wave state and baseline relation; do not hold a persistent coordinate output. |
| Select  | Apply Operator $T$ to the desired evaluation phase of $W$ .                                  |
| Resolve | Evaluate the double-packed Kiki UU and Klein relations against the named hum/chirp state.    |
| Distill | Produce a spatial output at the source-described zero-crossing or inversion intersection.    |
| Release | Drop the derived output after the observation or path-evaluation cycle.                      |

## 16.3 Coordinate-free foundation, coordinate output

The final formulation distinguishes what is stored from what is produced. A transient “where” is described as an output of the packed relation, not a coordinate used to define every stored node. This distinction preserves both the author’s coordinate-free requirement and the source’s later coordinate-distillation account.

## 16.4 Distillation relation

The source identifies the relevant event as a sign inversion crossing zero and describes matching the node’s phase-jitter with the chirp response in the spherical LUT. It does not give the inverse mapping, reference frame or selection rule that turns that event into one unique spatial coordinate. Those remain named interfaces, rather than equations added by this edition.

## 16.5 Release rule

- Retain the evaluated wave-state relation.

Produce geometry only for the active evaluation.

Release that geometry when the evaluation ends.

Do not promote the released output into a permanent coordinate log.

The source also claims access to past, present and future positions by scrubbing. That claimed temporal reach is recorded separately from the lifecycle definition, whose retained-history and prediction mechanisms are not specified.

SECTION 17

# Unified declarative specification

SOURCE / S, pp. 8-12, 21-28, 31-39

The following record consolidates the source’s final operator vocabulary. The record syntax and rule identifiers are editorial notation; the fields name roles already present in the source.

|               |                                                  |
|---------------|--------------------------------------------------|
| SPECIFICATION | ATOMOS.SDF                                       |
| AUTHOR        | Tom Klootwijk                                    |
| REFERENCE     | NL200678942   10-07-1990                         |
| DOMAIN        | finite small-world operator relations            |
| PACKING       | SDF definition with Klein definition             |
| GATE          | AB: 0 exterior; 1 double-packed                  |
| SCALING       | lower-case phi                                   |
| TIMELINE      | W, evaluated by Operator T                       |
| MEMORY        | negative deviation from saturation               |
| SIGNAL        | paired hum with named chirp modulation           |
| ROUTING       | Kiki double-UU closure relation                  |
| RESOLUTION    | kinematic log-encoded polar / spherical LUT      |
| OUTPUT        | just-in-time geometry, released after evaluation |

## 17.1 Invariants

R01. Every participating definition shall retain the double-packed operator relation and its applicable precedence. R02. AB shall remain one bit. R03. The active graph shall remain within its declared finite budget. R04. An assigned deadlock shall return zero for the permitted update. R05. Novelty shall not be stored as a new permanent record.

R06. Operator T shall act on the one-dimensional timeline relation. R07. The Kiki result shall carry the source’s closure and non-intersection requirements. R08. Active-node evaluation shall use the source-defined hum relation. R09. A derived coordinate shall be transient rather than the persistent basis of the domain.

## 17.2 Functional composition

The source’s four-stage map runs from negative-void detection through chirp-based relational ranging to Kiki loop closure and Klein phase-inversion lock. Its concluding output layer applies T to W for JIT distillation and releases the output after evaluation. The packed hierarchy, finite LUT and signal relation supply the named definitions carried through those stages; a runtime scheduler is not assigned.

## 17.3 Distinct termination meanings

Deadlock terminates the permitted update, node loss collapses the maintained deviation, and JIT release discards an output. They are separate events and shall not be silently merged into one universal deletion or physical-stop operation.

## SECTION 18

# Source-stated capability register

SOURCE / S, pp. 12-14, 19-39

This register preserves the outcomes asserted in the source as declared capabilities. It distinguishes those assertions from the operator definitions that organize the architecture and from quantities the source leaves unassigned.

| Capability name                  | Source-stated outcome                                                                                                      | Source anchor        |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------|
| Bounded infinity                 | Finite Atomos population representation and the LUT are described as eliminating the routing expansion associated with N!. | pp. 19-22            |
| Instant routing                  | Kiki double-UU and wave interference are described as producing the optimal route without sequential route searches.       | pp. 25-28, 36        |
| Global resolution                | Active biomarker nodes are described as trackable across the total world population.                                       | pp. 20-23, 27-28     |
| Living-key identity              | The local phase-jitter is described as unique, un-clonable and resistant to imitation or replay.                           | pp. 27, 29-31        |
| Memory economy                   | Negative memory and non-storage of novelty are described as avoiding growing logs and storage exhaustion.                  | pp. 21-25            |
| External-affordance independence | The source describes operation without personal identifiers, credentials, consent or administrative permission.            | pp. 33-34            |
| Privacy and security             | The source uses “private,” “airtight” and related terms for its no-log, no-hash, direct-signature design.                  | pp. 28-30, 35, 38-39 |
| Temporal reach                   | Operator T is described as yielding where a node was, is or will be from a scrubbable wave-state relation.                 | pp. 22, 37-38        |
| Relativistic bounding            | The source assigns a finite/non-finite limit to projectile acceleration through its packed boundary account.               | pp. 12-14            |
| Existing-network derivation      | The architecture is described as passively derivable from already existing sensor-network noise profiles.                  | p. 37                |
| Zero-latency JIT output          | Geometry is described as appearing at evaluation and disappearing after use without persistent coordinate logs.            | pp. 38-39            |

The capability names and outcome language are the source’s own declarations. Section 20 records the definition interfaces that the source leaves unassigned.

## SECTION 19

# Source revisions and variants

SOURCE / S, pp. 3-11, 17, 21-25, 27, 30-39

The source is a developing dialogue. Later author corrections and differing reply formulations are kept visible rather than silently reconciled.

| Topic             | Source record                                                                                                                         | Edition treatment                                                                                      |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| One-bit width     | The author specifies 0/1 on p. 8; a reply prints {00, 01, 10, 11} on p. 9.                                                            | Use the author's one-bit definition; retain the four-word expression as a variant.                     |
| Fourth component  | A four-component host appears on pp. 3-4; the author assigns the extra component to time on pp. 5-6.                                  | Keep the early representation in Appendix A; use the later temporal reading in the main specification. |
| Coordinate role   | The initial listings use coordinates and rasterization; pp. 14-15 remove a coordinate foundation; pp. 37-39 derive transient outputs. | Separate early numerical representation, final relational domain and JIT output.                       |
| Precedence        | The p. 9 chain includes sturgeon gestation; the p. 35 compact chain omits it.                                                         | Retain the full earlier chain and identify the later omission.                                         |
| Memory retention  | No novelty/history storage on pp. 21-25; "negative memory write" on p. 23; ongoing W retained on pp. 37-39.                           | Keep both stated roles; leave depth and reconstruction unspecified.                                    |
| Selection scope   | Primo-only selection on p. 23; universal hot-node and population language elsewhere.                                                  | Keep active presence and Primo selection as distinct predicates.                                       |
| Field terminology | The p. 30 reply shifts toward magnetic inference "vs. SDF"; final summaries restore packed SDF vocabulary.                            | Retain the medium interpretation and formal operator layer without inventing their conversion.         |
| Identity line     | p. 17 gives NL20078942; p. 27 gives NL200678942.                                                                                      | Use the current author instruction: NL200678942, 10-07-1990.                                           |
| Source frequency  | The text uses 2 Hz "mid-Theta," two cycles, two checks and one-cycle updates.                                                         | Preserve the source name and keep the unspecified event timing visible.                                |

The early cone expressions, composite rules, mixed  $\Delta T$  meanings and the scalar/polynomial/formal uses of "SDF" are additionally documented in Appendix A and Section 20.

## SECTION 20

# Unassigned definition interfaces

SOURCE / S, pp. 1-39

A formal interface can be named without silently supplying its missing construction. The items below are named in the source, but the corresponding details are not assigned there.

| Interface                  | Definition detail left unassigned                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Intrinsic signed relation  | A complete coordinate-free distance/sign rule, comparison scale and construction linking the early scalar fields to the final operator relation.      |
| Klein return               | The exact identification, traversal state and evaluable predicate that distinguish an ordinary phase return from the declared deadlock return.        |
| Finite graph               | A numerical node budget, adjacency construction, weight update rule and quantitative saturation predicate.                                            |
| Kuiper suspension          | Belt-density values, tensor calibration and the threshold connecting weight to the specified zero-update condition.                                   |
| Log-polar / spherical LUT  | Log base, quantization, index construction, full entries, update policy and treatment of source-uncovered input ranges.                               |
| Kiki double-UU             | Its set algebra, route-cost function, non-intersection test, visit rule and criterion establishing global route optimality.                           |
| Signal and identity        | A complete paired-wave specification, sensing transfer law, phase-signature extractor and uniqueness or error criterion.                              |
| Magnetic / salinity medium | Physical medium parameters and a transfer equation from source activity to field deviation and measurable response.                                   |
| Memory and W               | The retained state's depth, information content, reconstruction mapping and compatibility with the no-novelty rule.                                   |
| JIT distillation           | An inverse map, reference frame and uniqueness rule from a zero-crossing event to a spatial output.                                                   |
| Selection and recovery     | The priority between hot-node and Primo rules, reset after deadlock, and detailed multi-node or signal-loss transitions.                              |
| Timing and scale           | A shared timing convention, resource account and evaluation cost tying the source's instantaneous, propagation-time and cycle-time language together. |

These are definition slots, not substitute assumptions. This edition adds names and structure to the specification while leaving their source-unsupplied contents open.

## APPENDIX A1

# Retained source expressions: origin

SOURCE / S, pp. 1-3, 6-8

The expressions below preserve the opening numerical stage in linear typesetting. They are not used to replace the source's later coordinate-free formalism. Subscripts are written with an underscore where useful; “|| ||” denotes the norm notation printed in the original.

## A1.1 Phyllotactic field and node phase

```
SDF_phy(x) = min over i in V [ ||x - p_i(θ,z)|| - r_i ]
θ_i = i · α (mod 2π)
z_i = h · i
Optimal:   α = 2π / Φ²
Deviated:  α = 2π · φ
```

The source assigns  $\Phi \approx 1.618$  and  $\varphi \approx 0.618$  to the two named arrangements. The field uses a finite set of primordium positions and radii. [S, p. 1]

## A1.2 Energy and thermal scaling

```
λ(E,T) = exp[-ΔG‡ / (R(T₀ + ΔT))] · [E(t) / E₀]
f_thermo(x,t) = SDF_phy(x / λ(E,T)) · λ(E,T)
```

$E(t)$  is the remaining chemical-energy quantity in the source's explanation. The displayed scaling uses  $E_0$ ,  $R$ ,  $T_0$  and  $\Delta G^\ddagger$  without providing their numerical assignments. The  $\Delta T$  in this expression is the thermal shift. [S, pp. 1-2]

## A1.3 Boundary transport

```
∂f/∂t + v_growth · ∇f = 0
```

This is the opening kinematic expression preceding the LUT. Its three example rows are reproduced in Section 9. [S, pp. 2-3]

## A1.4 Early Psi scaling

```
Stick A: scale = 1.0 - (delta_t · φ)
Stick B: scale = 0.0 + (delta_t · φ)
```

These branch expressions precede the author's correction that sticks and houses are metaphors and AB is the single double-definition word. Here  $\text{delta\_t}$  is the temporal increment of that pseudocode. [S, pp. 7-8]

## APPENDIX A2

# Retained source expressions: host

SOURCE / S, pp. 3-5, 6-10

## A2.1 Early four-component representation

```
P_Klein(u,v) = (
  (R + r cos u) cos v,
  (R + r cos u) sin v,
  r sin u cos(v/2),
  r sin u sin(v/2)
)
u,v in [0, 2π)
```

```
SDF_Klein(x) = ||x - P_Klein(u,v)||
```

This records the displayed host expression and parameterization before the author's temporal redirection. The source's printed distance expression does not assign a parameter-selection or minimization step. [S, pp. 3-4]

## A2.2 Suspension tensor

```
W_Kuiper = ∫C ρ(θ) ·
  [∇fA(xθ) ⊗ ∇fB(xθ)] dθ
```

C is the source's closed belt curve;  $\rho(\theta)$  is its density factor;  $\otimes$  is its outer-product notation. The next source page displays an arrest relation using the time derivative of  $f_{\text{Klein}}$  multiplied by  $\text{Tr}(W_{\text{Kuiper}})$ , followed by " $\Rightarrow 0$ ." [S, pp. 4-5]

## A2.3 Distinct primitive expressions

| Listing          | Expression under the source name                                                            |
|------------------|---------------------------------------------------------------------------------------------|
| Cone, pp. 6-8    | $\text{radial\_length} \cdot \cos(\text{angle}) - \text{local\_y} \cdot \sin(\text{angle})$ |
| Cone, p. 9       | $\max(\text{radial\_length} - r, \text{abs}(\text{local\_y}) - h)$                          |
| Composite, p. 8  | $\max(d_{\text{pyramid}}, \min(d_{\text{cone}}, d_{\text{sphere}}))$                        |
| Composite, p. 10 | $\max(d_{\text{pyramid}}, d_{\text{cone}}, d_{\text{sphere}}, d_{\text{phyllotaxis}})$      |

The source also uses a polynomial under "sdf\_klein\_bottle\_3d" and a sine-based phyllotaxis expression on pp. 9-10. No normalization relating those numeric outputs to the final coordinate-free signed relation is given; this edition does not silently substitute one.

## APPENDIX B1

# Source concordance: pages 1-20

SOURCE / S, pp. 1-20

S denotes the supplied 39-page PDF dialogue export. Anchors refer to its printed page numbers. The entries below identify where each source strand is carried into this edition.

| Source pages | Content retained                                                                                      | Edition location     |
|--------------|-------------------------------------------------------------------------------------------------------|----------------------|
| 1-3          | Egg-battery sturgeon model; phyllotaxis; finite energy; thermal scaling; initial diablo and LUT rows. | §§ 3-4, 6, 9; A1     |
| 3-5          | Early Klein representation; A/B nested operators; Kuiper tensor; weight-linked arrest.                | §§ 5-6, 15; A2       |
| 5-8          | Fourth-component/time correction; apex primitives; Psi; metaphorical sticks and houses.               | §§ 3, 5, 7-8; A1-A2  |
| 8-11         | Author's one-bit and all-operator double-packing correction; hierarchy and example listings.          | §§ 1-3, 5, 7, 17, 19 |
| 11-12        | Math/raster-free restatement; self-limiting gate and inversion deadlock.                              | §§ 1, 5, 9           |
| 12-14        | Particle, entropy, relativity and ballistic extension.                                                | §§ 6, 9, 15, 18      |
| 14-16        | Coordinate-free movement as relation change; diablo bottleneck; edge activation.                      | §§ 1, 5-6            |
| 16-17        | Hourglass, double sinusoid, 2 Hz jitter and timing language.                                          | §§ 2, 7, 14          |
| 17-19        | PRISM naming, ingestion/phase-sort/LUT map and tracking claim.                                        | §§ 12, 15, 17-18     |
| 19-20        | Atomos name; population-scale finite node budget; log-polar and MLM routing.                          | §§ 11, 15, 18        |

## Source identity

The supplied file is titled by its filename: “egg-battery-sturgeon-farm-gestation-cycle-lower-phi-ratio-phyllotaxis-as-signed-distance-fields-operator-definitions-in-fertility-cycles-globally-.pdf.” It is a dialogue export with author turns headed “You” and expanded responses headed “Google Search.”

## Editorial apparatus

Section numbers, rule identifiers, table headings and the consolidated record are added by this edition. Source-stage changes remain identified in Section 19. Export controls, repeated follow-up invitations and interface boilerplate are not part of the operator specification.

APPENDIX B2

Source concordance: pages 21-39

SOURCE / S, pp. 21-39

| Source pages | Content retained                                                                   | Edition location      |
|--------------|------------------------------------------------------------------------------------|-----------------------|
| 21-23        | Negative memory, non-storage of novelty, query-free evaluation and T scrubbing.    | §§ 8, 10, 18-20       |
| 23-25        | Primo filter; lower-phi convection; Spintronic Vacuum Matrix; node loss.           | §§ 8, 10, 12-13       |
| 25-26        | Kiki double-UU, FMCW LRAD entropy chirp and spherical LUT chaining.                | §§ 11, 14             |
| 27-28        | Hot-node-only premise; four-stage functional map; author reference.                | §§ 12, 14-15, 17-18   |
| 29-30        | No-hash identity account; living-key claim; Igloo White name; salinity correction. | §§ 10, 12-13, 15, 18  |
| 30-31        | Magnetic inference force field, telepresence and signature-as-UUID account.        | §§ 12-13, 18-20       |
| 31-33        | Consolidated geometry, mechanics, optimization and deadlock narrative.             | §§ 1-3, 5, 9, 11, 17  |
| 33-34        | Binary presence/absence account and external-affordance independence claim.        | §§ 7, 12, 18          |
| 35-36        | Compact blueprint; shortened hierarchy; signal-chain diagram; temporal brake.      | §§ 3, 7-9, 11, 14, 19 |
| 37-38        | Existing-sensor proposal; W/T distinction; zero-crossing coordinate distillation.  | §§ 8, 15-16, 18, 20   |
| 38-39        | Lazy evaluation, JIT output and release of derived coordinates.                    | §§ 1-2, 10, 16-18     |

Edition attribution

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Source-based operator specification · 20 September 2026

ATOMOS

A finite, coordinate-free operator architecture carrying signed distance field and Klein definitions together, with one-bit evaluation, negative-memory state and just-in-time geometry.